

Lost Code, Found Culture; Emulation, Abandonware, and Preservation

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Over 37,200 video games that were made have been abandoned (myabandonware.com, 2026). They are not supported by the people who made them, the publishers that sold them, and oftentimes, even the console they can run on was abandoned, too. Preservation efforts are already underway, but so many games are already lost. Art is made to be enjoyed, and emulation allows this art to be enjoyed indefinitely. Emulation is widely frowned upon; a lot of people believe it is fully illegal, and lawsuits over the years don't help change that perspective, though most resolved as emulators being legal (Feng et.al. 2025). There are legal ways to emulate games, which will allow for easier preservation, sustaining the value of the secondary market, and giving freedom of choice to the player.

Benefits of Legal Emulation in Game Preservation

Legal Emulation is Preservation

One of the major caveats when looking at emulating games is that most people don't realize they still need to own the game, a physical copy of it, to legally emulate it. Most of the time, as long as an individual owns a physical copy, they can do whatever they want to it, with one major limitation. Redistribution of the code of a game from a single copy is illegal; however, an individual can decompile it from the original disc or cartridge and recompile it to work on a personal computer. There are a couple of advantages to doing this. A gamer can make a backup of the game, playable on either the original hardware or the computer. It also allows for alteration of the code, known as a ROMhack, which allows more replayability for titles. Decompiling and playing on a personal computer is a legal way to obtain the game's code, but it is still viewed as unethical by the game's developers, and protections are put in place to prevent

this action on newer games. According to Dalvi (2023), the legality of decompiling code has become more of an ethics issue:

In the United States, there is absolutely NO law that restricts decompiling software or copying it for personal use. Companies have tried to sue to stop it, but that's only civil, not criminal, and therefore not illegal, and they've only won when the content was given to an outside party from whom the companies did not receive payment, and that person has been shown to break the law (Section 9).

The issue of ethics surrounding decompilation is caused by the producers or publishers preventing their code from being distributed freely; distributing source code or ROMs is illegal (LegalClarity Team, 2026). Decompiling code to make a personal backup is legal, but the stigma of decompilation makes software that can decompile games nearly impossible to find.

Sustaining the Value of Collector Items

The secondary market for these games has always been a bit expensive. A copy of *The Legend of Zelda: Twilight Princess* costs \$174 today when it was \$50 at release (pricecharting.com, 2006-2026). These games are expensive and are only going to get more expensive as time goes on. The only legal way to play these games today is to own original hardware or emulate them, and the only legal way to emulate a game is to own a copy of the game and decompile it oneself. Obtaining a copy of these games is getting more and more difficult, due to the system as it currently is. Digital media such as a video game can be lost forever; however, there are new laws being proposed and put in place to protect these art mediums (Ward, 2026). Despite these setbacks, the only way to legally emulate any game now is to decompile it, which means an individual needs to own a copy of the game to emulate it.

Collectors and store owners need not worry about legal emulation; it supports the prices of the secondary market, ensuring the price of the games for years to come.

Freedom to Experience Differently

Emulating a game also adds a lot of replayability and even more freedom than a player would usually have, adding accessibility features like save-states, which give the ability to save the game at any point, and remappable controls or different controllers than usual. Perhaps the most extreme example of this is something known as a ROMhack which is a major change to the code of the game that allows for much more freedom of how the game is played (Cummins, 2025). One of the more popular types of ROMhacks is what's known as a randomizer, moving items that are needed for progression to random locations within the game. For example:

OoTMM, a combination randomizer that uses code from *The Legend of Zelda: Ocarina of Time* and *The Legend of Zelda: Majora's Mask* to make one massive game with items scattered throughout the worlds of both games (OoTMM, 2026).

Criticisms of Emulation and Their Refutation

Piracy Across the Web

Critics argue that even if emulators themselves are legal, they primarily exist to facilitate piracy (Perreault & Hooper, 2014). Emulation allegedly leads to widespread copyright infringement, harming developers and publishers financially. As stated by the LegalClarity team (2026), "Downloading a ROM from an unauthorized website or sharing one with others is copyright infringement. This is true even if the game is decades old, no longer sold, or seemingly forgotten" (para. 4). While it is true that there is a lot of piracy in the emulating community, the people who have the means to decompile the code themselves generally do. Decompiling the code for personal use is legal, even if game companies argue otherwise. Emulators themselves

are also, 100% legal given only a single caveat; the code for an emulation software must be different from the code of the console itself (U. S. Copyright Office, 2000). This is usually done through clean room reverse engineering, which is another way to say decompiling. Removing the negative connotation from emulation and decompilation would allow for each individual to have a permanent copy of a game, ensuring it's preservation.

Copyright v. Emulators

Some argue that publishers retain full ownership of their intellectual property and are under no obligation to preserve or rerelease old games. Allowing outside preservation undermines copyright law and corporate control (Arnall Golden Gregory LLP., 2024). Games are just as much a part of human culture as any other form of art.

Imagine if the only way to watch Titanic was to find a used VHS tape, and maintain your own vintage equipment so that you could still watch it. And what if no library, not even the Library of Congress, could do any better—they could keep and digitize that VHS of Titanic, but you'd have to go all the way there to watch it. It sounds crazy, but that's the reality we live in with video games, a \$180 billion industry, while the games and their history disappear. (Ennis. 2025. Introduction)

Video games are important to the entirety of human culture, but due to current laws, it is near impossible to preserve them. Even further some of the laws actively work against preservation efforts, which is why it is such a breath of fresh air to see new laws being made even now that protect both the consumer and the game itself. The producers themselves don't care about the games anymore, that's why they are considered abandonware in the first place. The secondary market is unaffected by copyright laws, and even new games have physical editions.

Against the Original Artwork

Some argue that modifying games through ROMhacks distorts the original artistic vision and should not be defended as preservation (Swalwell, n.d.). ROMhacks exist alongside the original games, the originals stand as a distinct copy while the ROMhack is something new based on the code of the original. Almost all of them are made from what is known as a patch file, which is used in coordination with an original copy of the source code to make a ROMhack. As one expert states “The community standard of distributing only patch files rather than complete modified ROMs, allowing users to apply changes to their legally owned game copies” (Cummins, 2025, Fig 9). This allows for individuals to have freedom in how they choose to experience the games, and adds to the longevity of the game, not only its preservation.

Preserving Games for Future Generations

In Retrospect

There are legal ways to emulate games: mainly decompiling an owned copy, allowing easier preservation by creating many copies of the games source code for everyone that owns it, sustaining the value of the secondary market, because the old games still need to be purchased, and giving freedom of choice to the player, either by playing the game itself or through a ROMhack. Emulation is the answer to an issue that has been created and necessitated by the laws that exist as they are. Games need to be preserved. They are as much a piece of art as any other form of media with one exception; games were made to be played, and as such, should remain in that way, even into preservation. Emulation is fully legal, but the laws are making it difficult to obtain a ROM for a game, even if a physical copy is owned; the issue is more one of ethics than legality. ROMs for abandoned games should be obtainable and playable, either by making decompilation more accessible, or making it possible to legally purchase ROMs for anyone who wants to play them, as that is the only proper way to preserve this artform.

References

- Arnall Golden Gregory LLP. (2024, January 18) *Decoding intellectual property law for video game developers*. <https://www.agg.com/news-insights/publications/decoding-intellectual-property-law-for-video-game-developers/>
- Cummins, D. (2025, October 19). *ROM Hacks 2026: Complete guide to game modification and preservation*. Bound by Flame. <https://boundbyflame.com/rom-hacks/>
- Dalvi, M. (2023, October 6) *Decompilers: ethical or unethical?* Medium <https://medium.com/@manas33/decompilers-ethical-or-unethical-b77c2c8854a8>
- Ennis, S. (2025, May 20). Preserving pixels: the DMCA and the quest to preserve video gaming's legacy. *Washington and Lee Law Review*, 82(2), 853-913. <https://scholarlycommons.law.wlu.edu/wlulr/vol82/iss2/8/>
- Feng, C. Xue, L. Cui. C. (2025, April 9). *Nintendo v. Yuzu: the legal boundaries of games console emulators*. Managing IP <https://www.managingip.com/article/2endvauyar5ncsxzj277k/sponsored-content/nintendo-v-yuzu-the-legal-boundaries-of-games-console-emulators>
- LegalClarity Team (2026, April 1). *Is emulating games legal? Emulators vs. ROMs* <https://legalclarity.org/is-emulating-games-legal-a-look-at-the-current-laws/>
- Myabandonware.com. (2026) *Website Introduction*. [Website] <https://www.myabandonware.com/>
- OoTMM, (2026) *OoTMM wiki* <https://ootmm.wiki.gg/>

Perreault & Hooper. (2014, November 22) *Emulators and ROMs: Legal and ethical implications of digital game piracy on the future of the games industry*

https://www.researchgate.net/publication/324866162_Emulators_and_ROMs_Legal_and_Ethical_Implications_of_Digital_Game_Piracy_on_the_Future_of_the_Games_Industry

Pricecharting.com. (2026) <https://www.pricecharting.com/game/gamecube/zelda-twilight-princess#completed-auctions-cib>

Swalwell, M. (n.d.) *Moving on from the original experience: Games history, preservation and presentation*. Flinders University. [\(PDF\) Moving on from the Original Experience: Games history, preservation and presentation](#)

U. S. Copyright Office. (2000) *Sony computer entm't, inc. v. Connectix corp*. U. S. Copyright Office Fair Use Index. <https://www.copyright.gov/fair-use/summaries/sony-connectix-9thcir2000.pdf>

Ward, (2026, March 19) *Assembly bill no. 1921*. California legislative information https://leginfo.legislature.ca.gov/faces/billTextClient.xhtml?bill_id=202520260AB1921